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Website Standards, Architecture & Development Life Cycle

Topics

1. The W3C
2. Web Design Principles
3. Web Design Standards
4. Website Architecture
5. Web Development Life Cycle

The World Wide Web Consortium (W3C)



- Established by Tim Berners-Lee in 1994
- Has more than 400 members from around the world
- It is a public-interest non-profit organization
- It has 12,000 developers worldwide participating in the standards development.
- It is an international multi-stakeholder community where member organizations, a full-time staff, and the public work together to develop open web standards.
- Purpose: To ensure the long-term growth of the Web.
- Activities: Develops standards and guidelines to help everyone build a web based on the principles of:
 1. Security
 2. Internationalization
 3. Privacy and
 4. Accessibility

Principles of Web Accessibility

- Web accessibility is about designing web sites, applications, technologies, tools, products and services in an inclusive manner (accessible by everyone), and thus lifting barriers to communication and interaction that many people face in the physical world.
- The Web is fundamentally designed to work for all people regardless of:
 - Hardware/device used - Websites should be responsive and usable on laptops, tablets, and mobile devices.
 - Software used - Content should work across multiple browsers (Chrome, Safari, Firefox, Edge/Explorer).
 - Language - Support for multiple languages (e.g. Eng, Russian, French) ensures inclusivity for global users.

Principles of Web Accessibility

- The Web is fundamentally designed to work for all people regardless of:
 - Location - Web should be available regardless of geographic location or Internet speed (e.g Manila, Mt. Apo, Ukraine)
 - Abilities/ disabilities:
 - Hearing → captions, transcripts for audio/video.
 - Movement → full keyboard navigation, switch input support.
 - Sight → alt text for images, screen reader compatibility, high contrast modes.
 - Cognitive → simple, consistent navigation, plain language, input help.

Principles of Web Accessibility

Web Content Accessibility Guidelines (WCAG) is a set of internationally recognized standards and guidelines to help web developers and designers create websites and web content that are accessible to everyone.

1. **Perceivable by all users** (including those with disabilities)

- Text Alternatives - Provide alternative text for non-text content
- Time-Based Media -- Provide captions, transcripts, and audio descriptions for multimedia content like videos and audio.
- Adaptable Content
 - Web content should be structured and flexible enough that it can be presented in different formats or through assistive technologies without losing its meaning, order, or relationships.

Principles of Web Accessibility

1. **Perceivable by all users** (including those with disabilities)

- Adaptable Content
 - Different users access content in different ways:
 - A blind user may use a **screen reader** that reads aloud text.
 - A user with low vision may need to **enlarge text** or use **high contrast mode**.
 - A user with dyslexia may benefit from **simplified layouts**.
 - A user on a mobile phone may see content **reflowed into a smaller screen**.

2. **Operable** - by a variety of input methods, including keyboard, mouse, voice recognition, and assistive technologies. Key guidelines under this principle include:

- Ensure that all functionality can be operated using a keyboard alone, without requiring a mouse.
- Create a clear and consistent navigation structure throughout the website to help users understand where they are and where they can go.

Principles of Web Accessibility

3. **Understandable** – Information and operation of the user interface must be understandable. This principle includes guidelines that focus on clarity and simplicity:
 - Readable Text -- text content is legible and understandable, with clear fonts, appropriate contrast, and understandable language.
 - Predictable Navigation -- Maintain consistency in design and layout, so users can predict how elements will behave.
 - Input Assistance -- Provide error messages, suggestions, and context for user input to help users correct errors.
4. **Robust** -- Content must be robust enough to work reliably with current and future technologies, including assistive technologies.
 - Compatibility with Assistive Technologies - ensure that web content is compatible with screen readers, braille displays, voice recognition software, and other assistive technologies.
 - Valid Code -- Write clean, well-structured code that adheres to web standards to enhance cross-browser and cross-platform compatibility.

Principles of Web Internationalization

- Web for All has been a fundamental concern and goal of the World Wide Web Consortium since the beginning,
- It is a natural requirement for Web-based applications to be accessible by anyone around the world,
- Not all are equal, people have different culture and practices
- A practice/content by a country is prohibited in other country
- focus on ensuring that the web is accessible and usable by people worldwide, regardless of language or cultural differences

Principles of Web Internationalization

What is Internationalization?

- Website internationalization is a term used to describe the process companies go through to make their websites more effective at reaching potential customers and business partners in other countries – trade.gov
- Internationalization (i18n) -- is designing and developing in a way that can be adapted for users from any culture, region, or language.
- It's essentially a Quality approach -- taking action early in the development cycle so that you avoid costly and sometimes prohibitive obstacles when it comes to rolling out your product to new marketplaces.

Principles of Web Internationalization

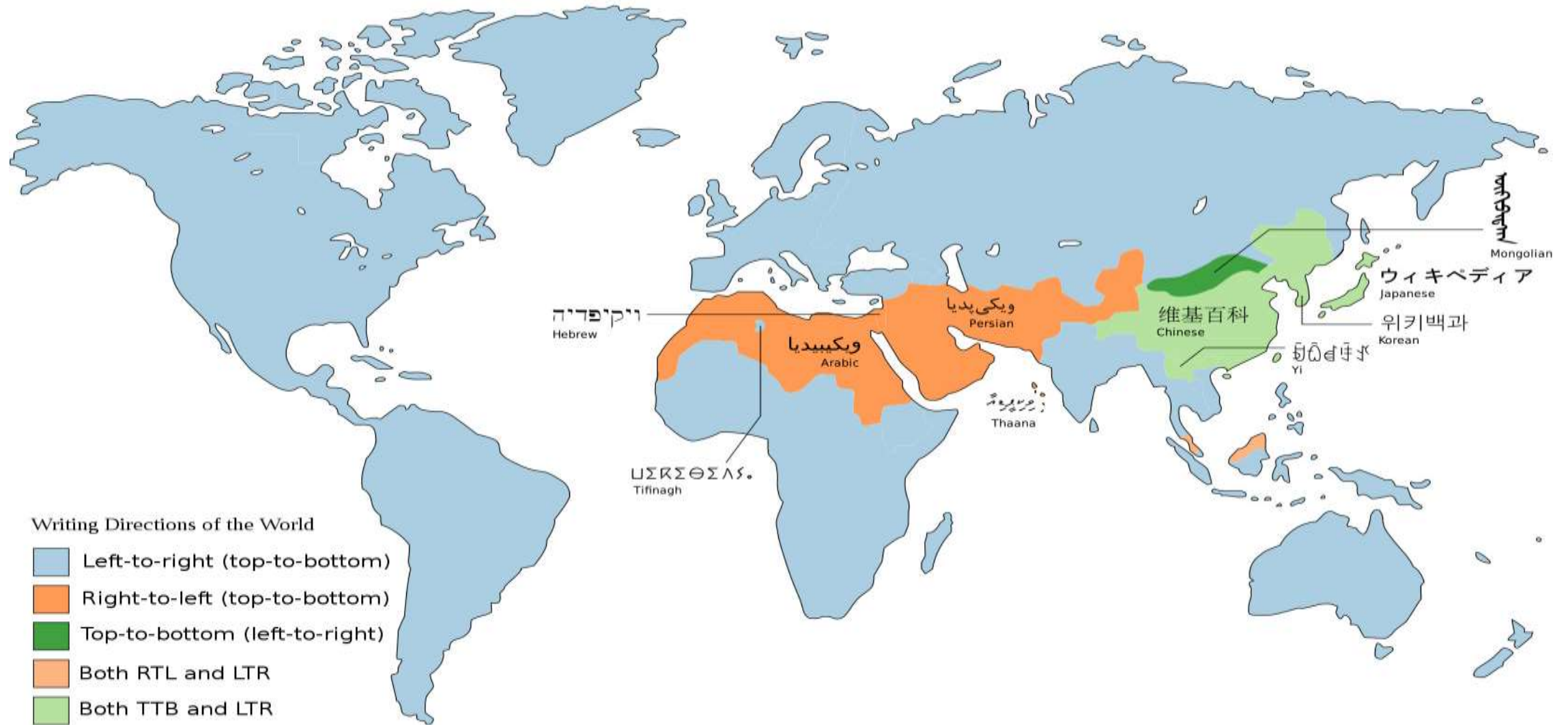
1. Universal Access

- accessible to people from diverse linguistic and cultural backgrounds
- promotes a web that is welcoming and inclusive to users from around the world.

2. Language Support

- The web should support a wide range of languages (3866 languages), scripts, and writing systems
- Ensures that web browsers can correctly display and process text in different scripts.
- Writing Direction:
 - Left to right: Most common
 - Right to left: Hebrew, Ancient Greek, Arabic, Persian (Farsi), Urdu, Kurdish, Pashto, etc
 - Top to bottom: Chinese, Mongolia, N. Korea, S. Korea, Japan

Principles of Web Internationalization



Source: <https://vividmaps.com/writing-directions-of-the-world/>

Principles of Web Internationalization

3. Character Encoding

- addresses character encoding issues to enable the representation and display of text in multiple languages.
- ensures that characters and symbols from different scripts are correctly encoded and rendered.

4. Language Tags

- used to identify the language of web content and resources
- emphasize the importance of accurately specifying the language of web content to enable proper language processing and user experience.

5. Cultural Adaptation

- The web should accommodate cultural differences in content presentation, such as date formats, calendars, and numbering systems.
- Cultural adaptation ensures that web content is culturally relevant and contextually appropriate.

Principles of Web Internationalization

6. Globalization of Software

- includes providing interfaces and tools that allow users to interact with web content in their preferred language and cultural context.

7. User-Friendly Interfaces

- Ensure that users can easily switch between languages and adapt web applications to their language and cultural preferences.

8. Locale Sensitivity

- considers locale-specific preferences, such as date and time formats, currency symbols, and measurement units.
- Web applications should be capable of adapting to these preferences based on the user's locale

Principles of Web Security

It protects both users and organizations from various online threats.

1. Authentication - Verify the identity of users and systems trying to access your web application.
 - Based on what you know -- password
 - Based on what you have— smart cards
 - Based on What you are - biometrics
 - Multi Factor Authentication - OTP
2. Authorization
 - implement proper authorization controls to ensure they only have access to the resources and functionalities they are allowed to use.

Principles of Web Security

3. Data Encryption

- secure data transmitted between the user's browser and your web server.
- encrypt sensitive data at rest using encryption algorithms to protect data stored on your server or database.

4. Input Validation

- Sanitize and validate all user inputs to prevent malicious data injection attacks
- Input validation should be applied both on the client and server sides.

5. Secure Development Practices

- Adhere to secure coding practices throughout the software development lifecycle.
- Employ security testing, code reviews, and penetration testing to identify and address vulnerabilities early in the development process

Principles of Web Security

6. Security Patching

- Keep all software, including web server software, frameworks, and libraries, up-to-date with the latest security patches.
- Vulnerabilities in these components can be exploited by attackers.

7. Least Privilege

- Limit the access and permissions of users and processes to the minimum necessary for them to perform their duties

8. Security Awareness

- Educate your development and operations teams about web security best practices.
- Promote a culture of security awareness within your organization.

9. Firewalls and Intrusion Detection: Use firewalls to filter incoming and outgoing traffic and intrusion detection systems (IDS) to identify and respond to suspicious activity.

Principles of Privacy

- The principle of Privacy emphasizes the need to protect users' personal information and data when they interact with websites and web services.
- These principles aim to establish a framework for responsible data handling and protection on the web, ensuring that users can trust the online environment with their personal information.
- These principles help guide the development of web standards and technologies that prioritize user privacy and data security.

Principles of Privacy

The principles of Privacy include the following:

1. Data Protection

- Websites and web services should implement measures to protect the privacy and security of users' personal data such as using encryption, secure authentication, and access controls to prevent unauthorized access or data breaches.

2. Transparency

- Websites should provide clear and concise privacy policies that inform users about the data they collect, how it will be used, and any third parties with whom it may be shared.

3. User Consent

- Websites should obtain explicit and informed consent from users before collecting or processing their personal data.
- Users should have the option to opt in or opt out of data collection practices.

Principles of Privacy

4. Data Minimization

- collect only the data that is necessary for the intended purpose and retain it for only as long as necessary.
- Unnecessary data collection should be avoided.
- For example: a website that sells products does not need to collect your birthdate or social security number.

5. User Control

- Users should have control over their personal data, including the ability to access, correct, delete, or export their data.
- Websites should provide mechanisms for users to exercise these rights.
- For example: users should be able to opt out of receiving marketing emails or having their data shared with third-party companies.

6. Cross-Border Data Transfer

- When personal data is transferred across international borders, websites and web services should comply with applicable data protection laws and regulations to ensure the privacy and security of user data.

Principles of Privacy

7. Privacy by Design

- Privacy considerations should be integrated into the design and development of websites and web services from the outset, rather than being added as an afterthought.

8. User Education

- Websites should educate users about privacy risks and best practices for protecting their personal information online.

9. Accountability

- Websites and organizations that handle user data should be accountable for their data protection practices and should have mechanisms in place for addressing data breaches and privacy violations.

Website Design Standards

- are guidelines, specifications and best practices that ensure websites are
 - functional,
 - accessible,
 - user-friendly.
 - good quality
 - Consistency
 - Accessible
 - and interoperability of websites across different platforms and browsers.
- They are the building blocks of a consistent digitally connected world.
- Guide the developer in making websites accessible to people with disabilities including providing alternative text for images, ensuring keyboard navigation, and avoiding content that could trigger seizures.

Benefits of adhering to standards:

- helps create a better user experience and
- simplifies the maintenance and future development of websites.
- help developers and designers create websites that are more reliable, accessible, and effective in delivering content and services to users.

Key Aspects of Website Standards

1. Technical Standards

- **HTML/CSS**

- Use of valid and semantic HTML for structuring content, and CSS for styling.
- Adherence to W3C standards ensures compatibility and accessibility.
- CSS is used to control the layout, styling, and presentation of web content.
- Web designers should adhere to CSS standards, including using external stylesheets, [avoiding inline styles](#), and organizing CSS rules efficiently.

- **JavaScript**

- Implement JavaScript in a way that is efficient and does not impede performance.
- Ensure compatibility with different browsers and devices.

Key Aspects of Website Standards

1. Technical Standards

- **Responsive Web Design**

- Browsing devices have different screen sizes, and other devices allow changes in screen orientation
- Websites should adapt to various screen sizes and orientations, providing an optimal user experience on smartphones, tablets, and desktops
- Use techniques like flexible grids and media queries to ensure websites work well on various devices and screen sizes.

- **Performance Optimization** - refers to website/ web page loading time

- Improve loading time by minimizing HTTP requests, compress files, and
- optimize images to improve load times and overall performance,
- minification (remove unnecessary characters) of CSS and JavaScript files, and the
- use of Content Delivery Networks (CDNs)
- key performance metrics: page speed, load times, and user engagement is part of web standards

Key Aspects of Website Standards

2. Accessibility Standards

- Compliance with the Web Content Accessibility Guidelines (WCAG) ensures that websites are usable by people with various disabilities
- This includes using alt text for images, ensuring sufficient color contrast, and providing keyboard navigability.
- Use HTML elements according to their intended purpose (e.g., <header>, <nav>, <article>, <footer>) to improve accessibility and SEO.
- providing alternative text for images (for visually impaired users), ensuring keyboard navigation, and avoiding content that could trigger seizures (for users with epilepsy).
 - minimize flashing and blinking, test for seizure risk, avoiding seizure-inducing content, provide Warnings

Key Aspects of Website Standards

3. Usability

- Focusing on user-centered design principles
- Design clear and intuitive navigation to help users find information easily. This includes logical menu structures and breadcrumb trails.
- Use legible fonts, appropriate font sizes, and sufficient line spacing. Organize content with headings and subheadings to enhance readability.
- Ensure buttons, links, and forms are easily clickable and provide feedback to users.

4. Design Consistency

- Visual Design - Maintain a cohesive visual style throughout the site, including color schemes, typography, and layout. Consistency improves user experience and brand recognition.
- Branding -- Integrate branding elements consistently, including logos, colors, and messaging, to reinforce brand identity.

Key Aspects of Website Standards

5. Security Standards

- Implementing security standards is crucial to protect websites and user data.
- use HTTPS to encrypt data transmission, protecting against common vulnerabilities like SQL injection and cross-site scripting (XSS), and keeping software and plugins up to date.
- Follow best data protection practices for storing and handling user data, including encryption and compliance with privacy laws.

6. Search Engine Optimization (SEO)

- Create clean URLs, providing meaningful metadata, and ensuring that websites are easily indexable by search engines
- Use relevant meta tags, such as titles and descriptions, to improve search engine visibility.
- Optimize content w/ relevant keywords & ensure proper use of headings (e.g. h1, h2).
- Create a logical site structure with a sitemap and proper internal linking to enhance search engine indexing.

Key Aspects of Website Standards

7. Cross-Browser Compatibility

- Websites should be designed and tested to work consistently across different web browsers, such as Google Chrome, Mozilla Firefox, Firefox, Safari, etc.
- Cross-browser compatibility ensures a broad user reach.

8. Content Strategy

- Provide Quality Content --- valuable, relevant, and up-to-date content that meets the needs of your target audience.
- Organize Content - Structure content in a way that is easy to scan and understand, using headings, lists, and clear language.

9. Legal and Ethical Considerations

- Privacy Policies -- Include clear privacy policies and terms of service, especially if collecting user data.
- Copyright Compliance -- Ensure that all content, including images and text, is properly licensed or owned.

Website Development

Technical Standards, Accessibility Standards, Usability Standards, Security, SEO, Cross-border, Content Strategy, Legal and Ethical Considerations

Principles of
Accessibility

Principles of
Security

Internatio-
nalization

Principles of
Privacy

Website Development Life Cycle (WDLC)

The WDLC is a structured approach to building a website, from initial planning to deployment and maintenance.

Phases in Website Development

1. Planning and Analysis
2. Design
3. Development
4. Testing
5. Deployment
6. Maintenance

Stages of the Web Development



WDLC: Planning and Analysis

1. Requirements Gathering
 - Understanding the client's needs, goals, and target audience.
 - Determine the goals, objectives, and functionalities of the website.
 - Analyze target audience, competitors, and industry trends.
 2. Feasibility Study -- Assessing technical and financial feasibility.
 3. Site Map -- Creating a visual representation of the website's structure and navigation.
 4. Wireframing -- Designing basic layouts and content placement.
- **Goal**: gain a deep understanding of the problem that a website is trying to solve so that the best solution can be found.

WDLC: Design

1. Develop Visual Design -- Create mockups and prototypes to visualize the website's appearance, including color schemes, typography, and imagery.
2. User Interface (UI) Design -- Designing the website's interface for optimal user experience.
3. User Experience (UX) Design -- Ensuring the website is easy to use and navigate.

WDLC: Development

1. Choose technology stack - Select programming languages, frameworks, and tools.
2. Front-End Development -- Building the website's visible interface using HTML, CSS, and JavaScript.
3. Back-End Development -- Build the server-side logic and database interactions.
4. Database Development -- Creating and managing the database to store and retrieve data.

WDLC: Testing & Review

Test the website comprehensively to iron out any bugs or errors before it goes live.

Categories of testing:

1. **Functional testing** -- focuses on verifying that the website performs its intended tasks.
 - Quality Assurance (QA) Testing -- Identifying and fixing bugs and errors in the website.
2. **Non-functional testing** -- assesses the website's performance, security, and usability.
 - Browser Compatibility Testing -- Ensuring the website works across different browsers and devices.
 - Usability Testing -- Evaluating the website's ease of use and effectiveness.

WDLC: Deployment

1. Publishing the Website -- Uploading the website's files to a web server.
2. Domain Name Registration -- Acquiring a domain name for the website's address (e.g. godaddy.com, vercel.com).
3. Hosting -- Selecting a web hosting provider to store the website's files (e.g. vercel.com, Netlify.com, render.com, github.com).

WDLC: Maintenance

Website maintenance is integral to the web development process for ensuring optimum performance and continuous upgrades based on evolving requirements.

1. Updates and Upgrades -- Keeping the website's software and content up-to-date. Implement new features, and bug fixes.
2. Security
 - Protecting the website from cyber threats
 - check for security vulnerabilities.
 - implement security patches

WDLC: Maintenance

3. Performance Optimization

- Ensuring the website loads quickly and efficiently.
- Track website traffic, speed, and uptime.
- ensure that your website remains relevant, updated, accessible, and optimal at all times for users.

4. Technical Support

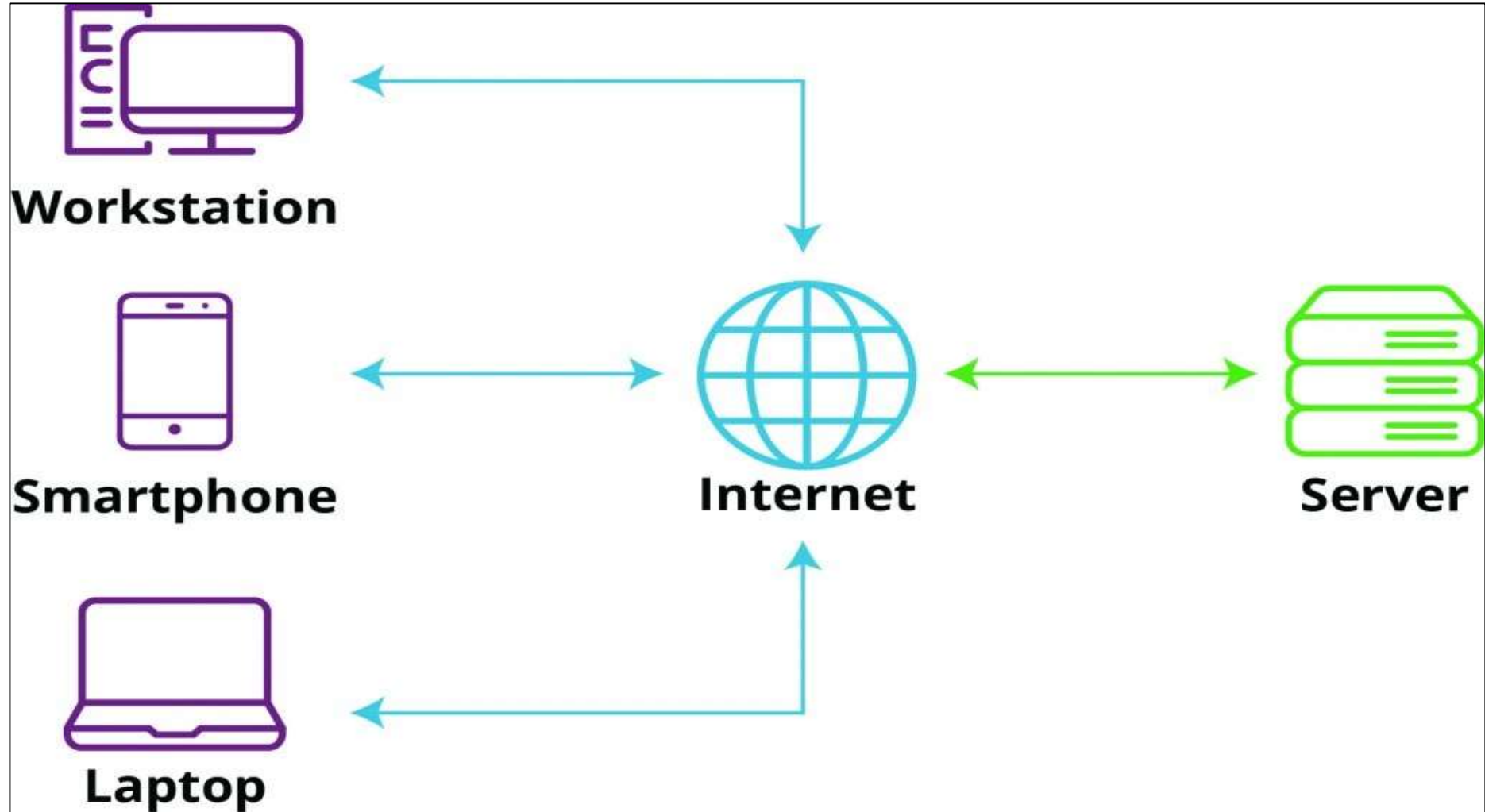
- Providing assistance to users and troubleshooting issues.
- Respond to user inquiries and suggestions

Maintenance is an indispensable phase of web development, so it's essential to allocate sufficient time and resources for maintenance as part of your overall web development strategy.

Website Architecture

- Refers to the structure and framework that determines how a website functions.
- It involves planning and organizing the various components of a website, including its:
 1. **Content** - refers to the text, images, videos, and other media that make up the website's content.
 2. **Functionality** - encompasses the actions and behaviors a website can perform, such as user interaction, data processing, and communication. It's often implemented using programming languages and frameworks.
 3. **Infrastructure** - refers to the hardware and software components that support the website, including servers, databases, and networks.

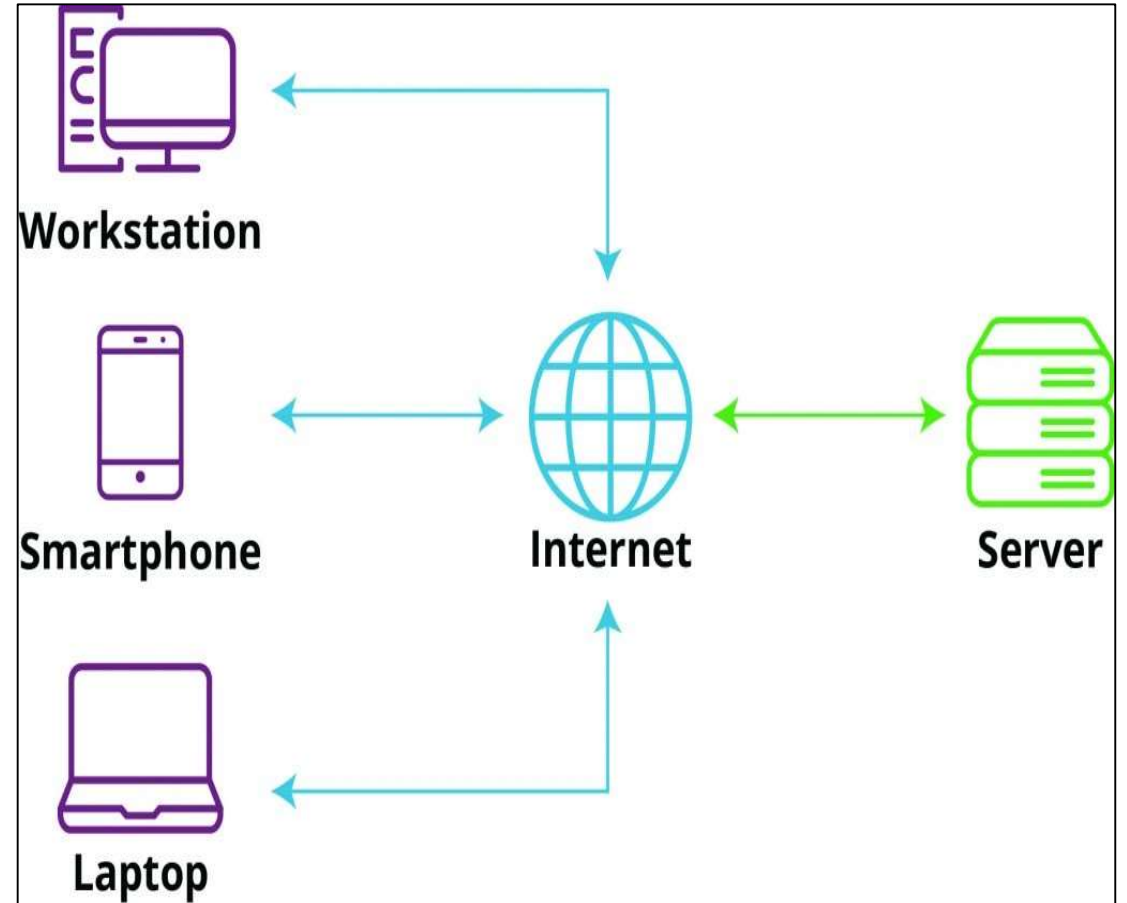
Overview of Client-Server Architecture



Client-server architecture

Client-server architecture is a type of computer network where multiple clients request and receive files and services from a centralized server over a local or internet connection.

Client-Server Architecture is a fundamental model used in **web** and networked systems where tasks and resources are divided between **clients** (users) and **servers** (providers of services/data).



<https://www.liquidweb.com/blog/client-server-architecture/>

Software organization in distributed systems

1. Clients

- are programs that use services that other programs provide
- typically handle user interactions
- often request data or initiate some data modification on behalf of a user.
- Example:
 - a client can provide a form onto which a user (a person working at a data entry terminal, for example) can enter orders for a product.
 - The client sends this order information to the server, which checks the product database and performs tasks that are needed for billing and shipping.
 - Typically, multiple clients use a single server.

Software organization in distributed systems

2. Servers

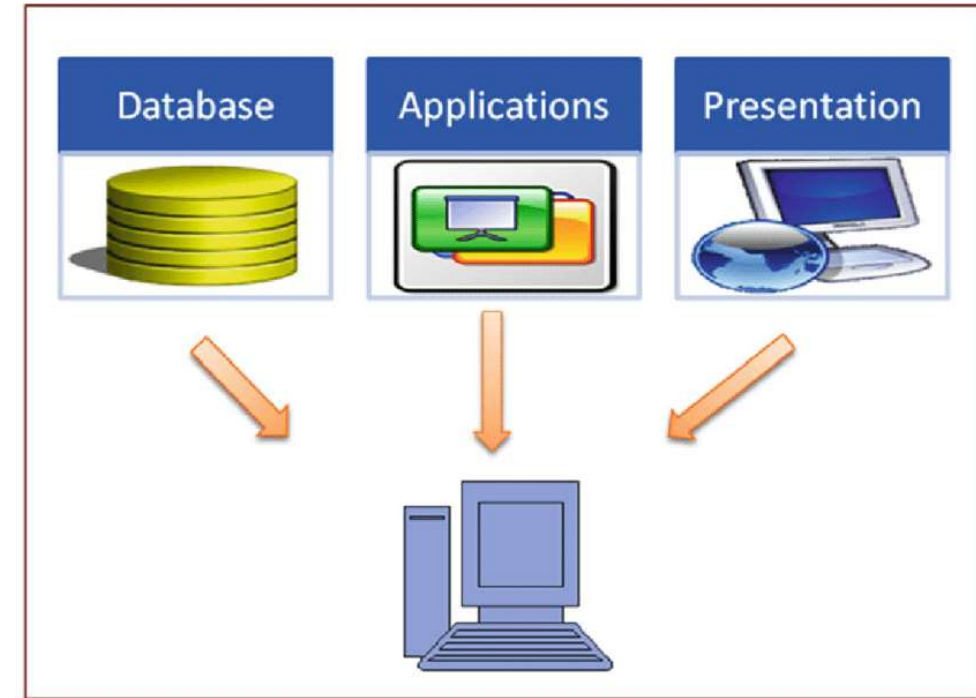
- are programs that provide the services.
- The server functions often require some resource management, in which a server synchronizes and manages access to the resource, and responds to client requests with either data or status information.
- Clients can also access several different servers, and the servers themselves can act as clients to other servers.
- Exactly how the functions are distributed across servers is an application design decision.
- For example:
 - a single server could provide all the services that a client needs, or the client could access multiple servers to perform different requests. The application designer must consider items such as scalability, location, and security.
 - are the clients and servers local or is the application distributed over a wide geographic area?
 - Do the servers need to be on machines that are physically secure?

Type of client-server architecture

Single-Tier/One-tier/ Monolithic Architecture

- the simplest type of software architecture where all components of a software application reside on a single server/machine:
 - User Interface / Presentation layer
 - business logic
 - data layer

Component	Location
User interface (UI)	In the same program
Business logic	In the same program
Data storage	Often local (e.g., file or local DB)

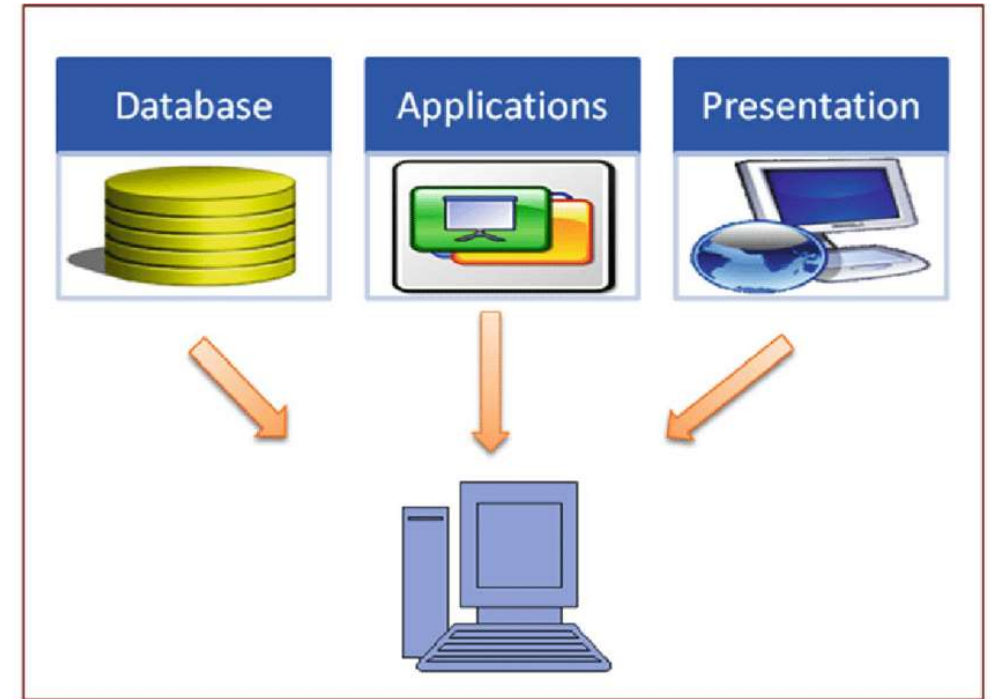


Type of client-server architecture

Single-Tier/One-tier/ Monolithic Architecture

Pros: Simplicity in design and deployment, easy to test.

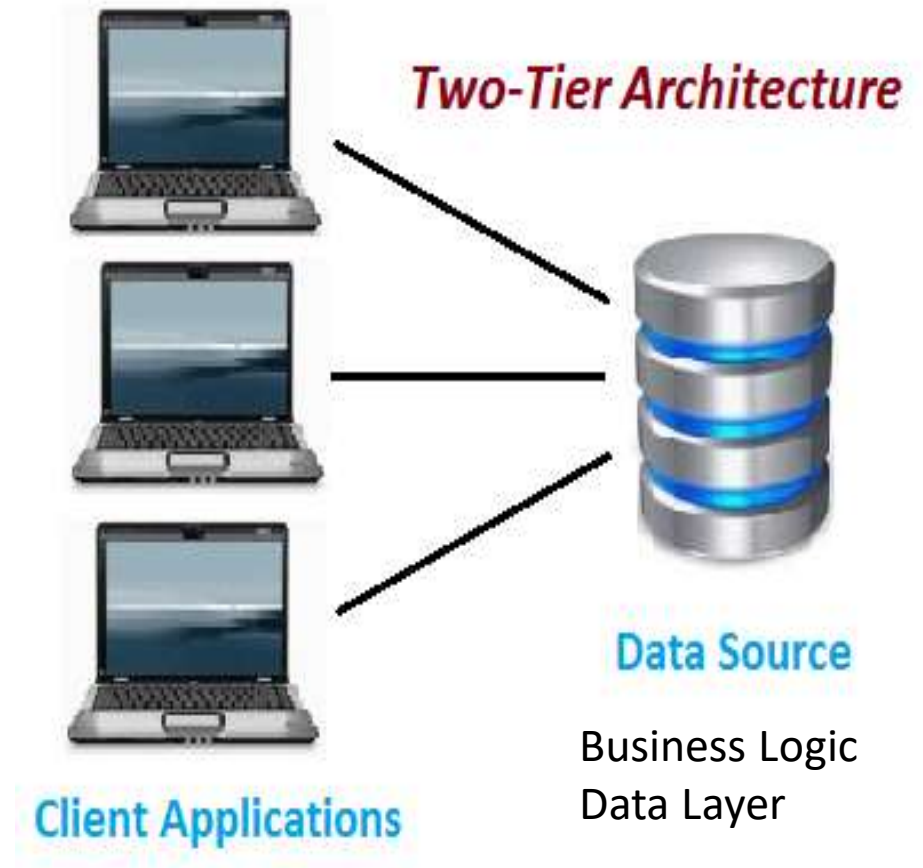
Cons: Scalability issues, difficulty in making changes, and potential for a single point of failure.



Type of client-server architecture

Two-Tier Architecture

- A two-tier architecture is a software architecture in which a:
 - **Presentation layer (Client Tier)** or interface runs on a client, and
 - Data Tier -- The **business logic application** and **data layer** or data structure gets stored on a server.
- Separating these components into different locations represents a two-tier architecture, as opposed to a single-tier architecture.

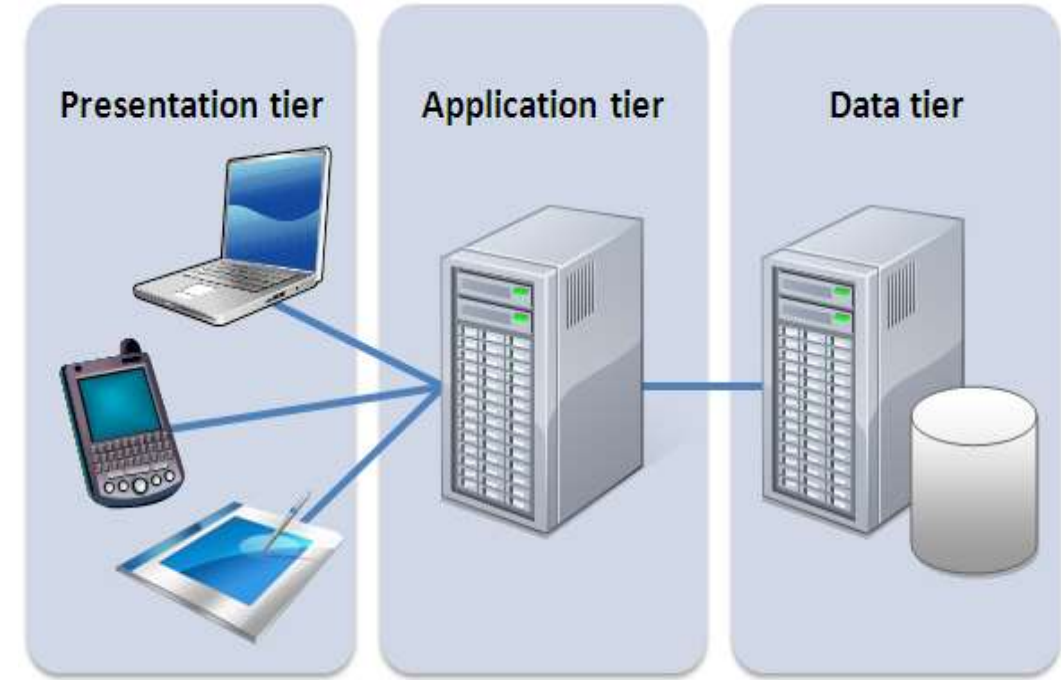


Type of client-server architecture

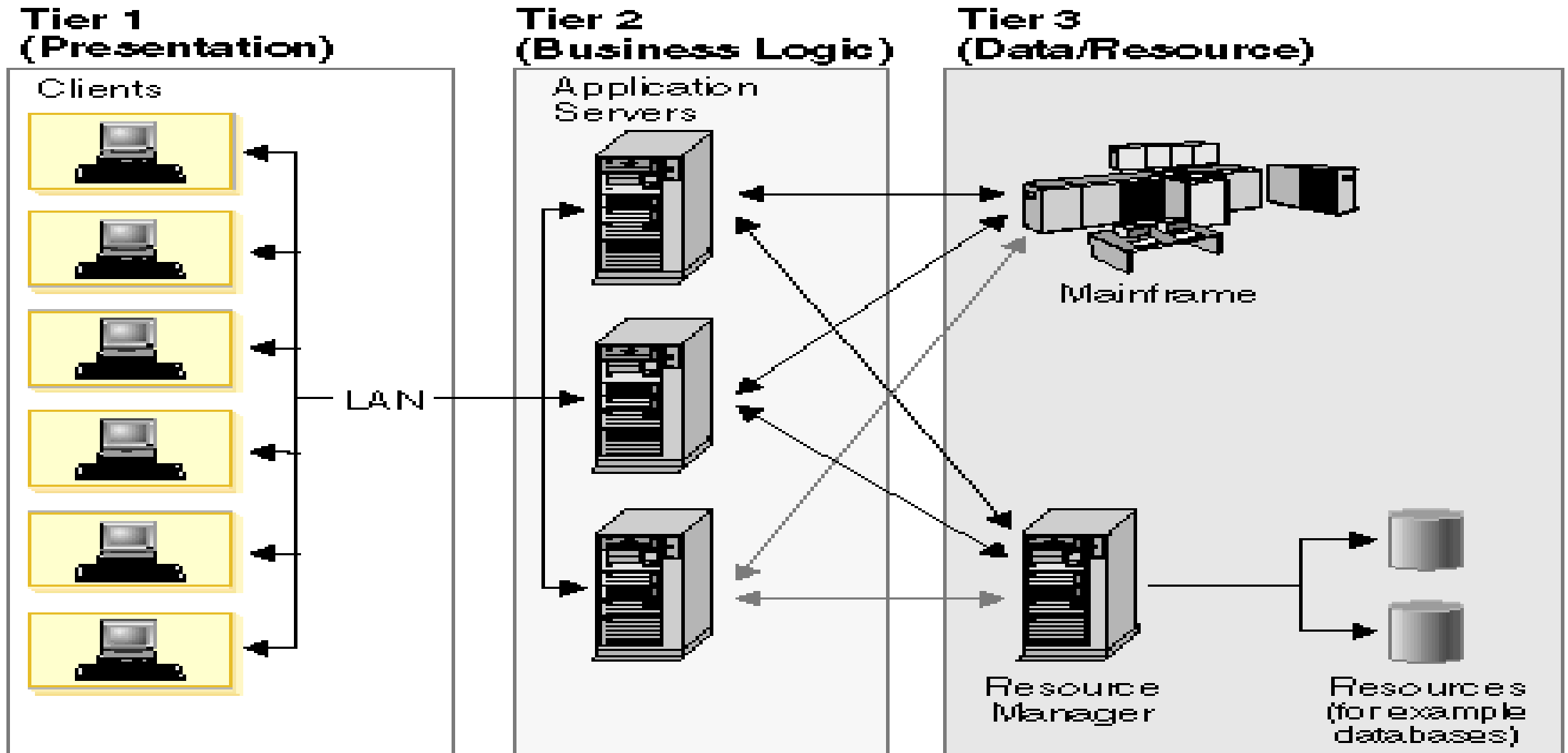
Three-Tier Architecture

Objectives: separate the users' view(s) of the database from the way that it is physically represented.

- a. Presentation Layer - Handles user interface and interaction.
- b. Application Layer - Processes business logic and data.
- c. Data Access Layer - Interacts with databases and other data sources.



Source: <https://managementmania.com/>

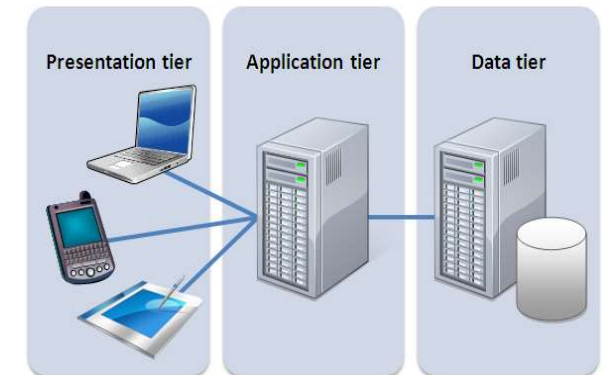


Source: www.ibm.com

Three-Tier Architecture

1. A **Presentation Layer** (front end, UI, or Client)

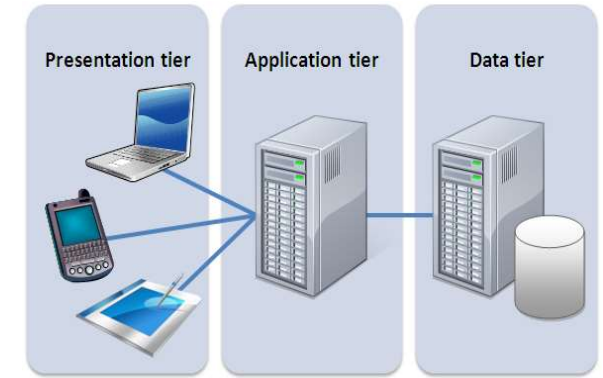
- the part of the application which is visible to the user;
- that sends content to browsers in the form of HTML/ JS/ CSS.
- it enables the input of requirements (e.g. filling up forms) and the presentation of results (e.g. viewing screen/ printing a report).
- It is dependent on the platform (e.g. [web applications](#), Windows applications, Android applications, etc.).
- This might leverage/use frameworks like React, Angular, Ember, Aurora, etc.



Three-Tier Architecture

2. An **Application Layer** or **Business Logic Tier**

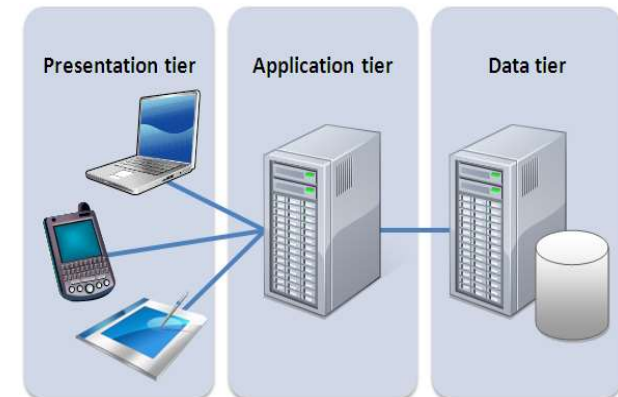
- Also known as the application server
- It comprises all the computation that is defined by all specific rules (e.g. customer discount)
- that uses an application server and processes the business logic for the application. This might be written in C#, Java, C++, Python, Ruby, etc.
- Serves as the middle layer of the model or 3-tier architecture
- it assures the calculations and operations performed between input-output requirements and data



Three-Tier Architecture

3. A Data Layer, Data Tier, or Back end

- Responsible for data management
- Provides backup and recovery services and transaction management
- it has a database management system (DBMS) that provides access to application data. This could be MSSQL, MySQL, or PostgreSQL, etc.
- it ensures all operations with data i.e. DBMS and basic data-base operations for functional storage, selection, aggregation, processing, integrity, and data audit.



Advantages of three-tier architectures

- **Scalability** – Middle tier can be used to reduce the load on a database server by using a transaction processing (TP) monitor to reduce the number of connections to a server, and additional application servers can be added to distribute application processing .
- **Technological flexibility**
 - Easier to change DBMS engines
 - middle tier can be moved to a different platform.
 - Simplified presentation interfaces make it easier to implement new interfaces.
- **Long-term cost reduction** – Use of off-the-shelf components or services in the middle tier can reduce costs, as can substitution of modules within an application rather than a whole application.

Advantages of three-tier architectures

- **Better match of systems to business needs** – New modules can be built to support specific business needs rather than building more general, complete applications
- **Improved customer service**
 - Multiple interfaces on different clients can access the same business process
 - Competitive advantage – Ability to react to business changes quickly by changing small modules of code rather than entire applications

Factors to Consider in Website Architecture

1. Scalability -- the ability to handle increased traffic and data volumes.
2. Performance -- how quickly the website loads and responds to user interactions.
3. Security -- protecting the website and its data from vulnerabilities and attacks.
4. Maintainability -- the ease of making changes and updates to the website.
5. Cost -- the overall cost of building, maintaining, and operating the website.



Thank you!

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2. <https://www.webfx.com/blog/web-design/best-wireframe-tools/>
3. <https://www.youtube.com/watch?v=qr0ujkLLgmE>
4. <https://www.youtube.com/watch?v=5nlcEgJGls8>
5. <https://www.youtube.com/watch?v=20SHvU2PKsM>